



ROYAL HOLLOWAY UNIVERSITY OF LONDON

Application of Machine Learning Methods to the Detection of Early Warning Signals of Atlantic Meridional Overturning Circulation Weakening

Nathan Burge

Supervised by: Prof. Veronique Boisvert

Department of Physics, RHUL
July 2026

Abstract

The Atlantic Meridional Overturning Circulation (AMOC) is a vital component of the Earth’s climate system but recent evidence suggests it may be weakening. Detection of early warning signals of a weakening AMOC remains a scientific challenge. This project explores the application of machine learning methods to identify potential early warning signals using ocean and atmospheric reanalysis datasets. Statistical analysis and machine learning methods are applied to historical climate variables to identify whether changes in predictive performance can evidence evolving AMOC behaviour.

Contents

1	Introduction and Background	2
1.1	The Atlantic Meridional Overturning Circulation	2
1.2	Evidence for AMOC Weakening	3
1.3	Tipping Points	5
1.4	Early Warning Signals	6
1.4.1	Critical Slowing Down and Statistical Indicators	7
1.4.2	Physics-Based Early Warning Signals	7
1.5	Machine Learning for Climate Prediction	8
1.5.1	Classification	8
1.5.2	Convolutional Neural Networks	9
1.6	Long Short-Term Memory Networks	9
1.6.1	Convolutional LSTM Networks	10
1.7	Research Objectives	10
2	Datasets and Methodology	12
2.1	Study Region	12
2.2	Datasets	12
2.2.1	ORAS5 Ocean Reanalysis	12
2.2.2	ERA5 Atmospheric Reanalysis	12
2.2.3	Variables Used	12
2.3	Data Pre-processing	12
2.3.1	Spatial Subsetting	12
2.3.2	Re-gridding and Combination	12
2.3.3	Normalisation	12
2.3.4	Training, Validation and Test Sets	12
2.4	Inflection Point Study	12
2.5	ConvLSTM Architecture	12
2.6	Model Training	12
2.7	Performance Metrics	12
2.8	Early Warning Signal Analysis	12
3	Model Performance	12
4	Conclusion	12
A	Additional Figures	12

1 Introduction and Background

1.1 The Atlantic Meridional Overturning Circulation

The Atlantic Meridional Overturning Circulation (AMOC) is one of the most important components of the Earth's climate system. It consists of a series of ocean currents that transport warm, saline surface waters northward through the Atlantic Ocean, while returning colder, denser waters southward at depth. This circulation plays a vital role in the regulation of regional and global climate conditions via the redistribution of enormous quantities of heat, freshwater and carbon around the globe. Approximately 1 petawatt (10^{15}Js^{-1}) of heat is transported northward across the Atlantic by the AMOC making it a major contributor to the relatively mild climate experienced across north-western Europe [?, 1].

The AMOC is a subsection of the global thermohaline circulation (illustrated in figure 1a), often referred to as the "global ocean conveyor belt" [1], which is driven by differences in seawater density. This seawater density is controlled primarily by temperature and salinity, with cold, salty water being denser than warm or fresh water. Surface water originating in the tropics is warmed by incident solar radiation and transported northward by the AMOC and Gulf Stream. Throughout its journey, this surface water releases heat to the atmosphere (particularly over the North Atlantic) causing it to cool. Simultaneously, evaporation increases its salinity. These processes work in combination to increase water density sufficiently for deep convection to occur in the Labrador Sea, Irminger Sea and Nordic Seas, where dense water sinks to form the North Atlantic Deep Water (NADW) [1, 2]. This deep water subsequently flows southward before eventually returning to the surface elsewhere through the upwelling process, completing the overturning circulation. Figure 1b shows the AMOC component of the global thermohaline circulation.

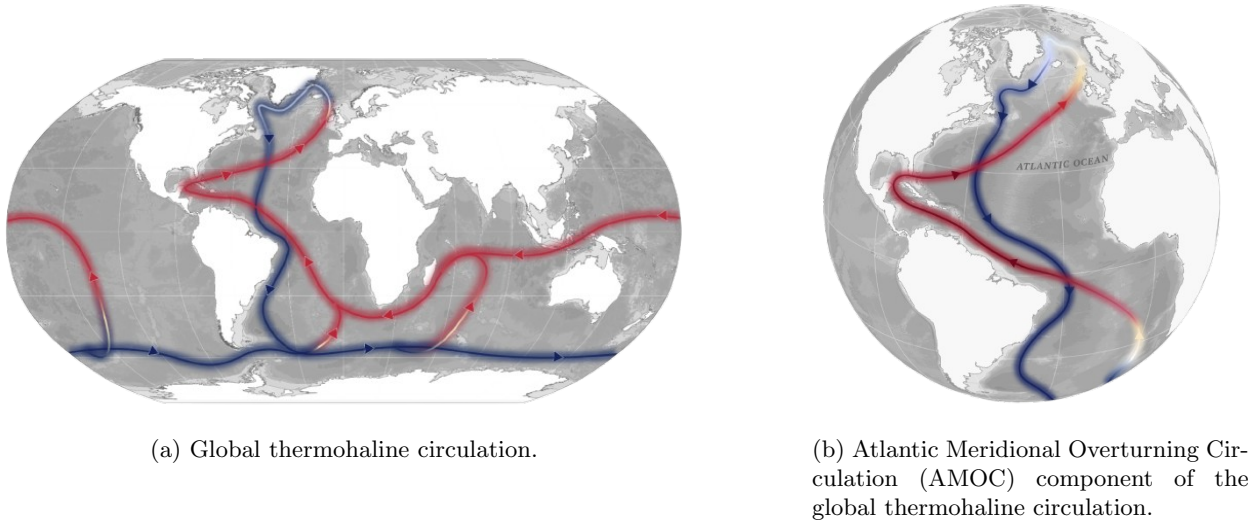


Figure 1: Schematics illustrating (a) the global thermohaline circulation and (b) the Atlantic Meridional Overturning Circulation. Adapted from [16].

The AMOC influences the global climate on timescales ranging from decades to millennia via heat transport, ocean ventilation and carbon storage. By transporting warm water northwards, it moderates winter temperatures across western Europe and affects atmospheric circulation patterns, storm tracks and precipitation. Beyond temperature regulation the circulation redistributes nutrients and oxygen, contributing to marine ecosystems, while also acting as a major component of the global carbon cycle through the storage and transport of dissolved carbon. As a result, changes in AMOC strength can have widespread implications [1, 2].

Paleoclimatic evidence demonstrates that the AMOC has not remained constant throughout Earth's history. During the last glacial period, multiple abrupt climate transitions, known as Dansgaard-Oeschger events, were associated with rapid shifts between strong and weak AMOC modes. Geological records indicate that these transitions occurred over relatively short timescales and were accompanied by changes in

temperature exceeding 5–10°C across Greenland, alongside changes in precipitation patterns, sea ice and climate on a global scale. These historical transitions demonstrate that the AMOC can possess multiple stable states and, when critical thresholds are exceeded, undergo abrupt and rapid restructuring [1, 5].

Future climate projections consistently suggest that anthropogenic climate change will weaken the AMOC. Increasing atmospheric temperatures lead to increasing freshwater forcing of the North Atlantic via enhanced precipitation and accelerated melting of the Greenland Ice Sheet. This freshwater introduction reduces surface salinity and therefore density, inhibiting sinking and the formation of deep water and weakening the overturning circulation. Although most current climate models predict a gradual reduction of AMOC strength in the twenty-first century rather than a complete shutdown, uncertainty remains regarding both the magnitude of this weakening and the possibility of abrupt transitions (a result of known biases in many climate models) [2–4].

Given the important role the AMOC plays in thermal regulation of the Northern Hemisphere climate and past evidence that it may exhibit non-linear behaviour, understanding whether the AMOC is approaching a critical transition has become one of the most important questions in recent climate study. The detection of reliable early warning signals of future weakening or collapse therefore represents a significant scientific challenge, with major implications regarding climate prediction and mitigation [5, 6].

1.2 Evidence for AMOC Weakening

Since the late twentieth century, increasing evidence suggests that the AMOC is weakening in response to anthropogenic climate change. Despite direct AMOC observations being limited, a combination of instrumental measurements, ocean reanalysis datasets, paleoclimate reconstructions and climate model simulations all indicate a decline in AMOC strength. Significant uncertainty remains regarding the timing and magnitude of this weakening. However, it is broadly agreed upon that continued global warming is expected to reduce AMOC strength throughout the twenty-first century [2, 3, 7].

A major difficulty regarding the assessment of long-term AMOC behaviour is the lack of continuous direct observations. Comprehensive, direct measurements of the AMOC have only been available since 2004 through the RAPID-MOCHA array (located at approximately 26.5°N in the Atlantic Ocean). While observations from this array have provided an unprecedented view of AMOC variability, the relatively short duration of the record results in inter-annual and decadal variability dominating the data, making long-term climate trends difficult to distinguish. Nonetheless, RAPID observations have still demonstrated substantial variability in AMOC strength and revealed periods of anomalously weak circulation [6, 7].

In order to overcome these limitations, researchers have developed methods of reconstructing historical AMOC strength indirectly using observable "fingerprints". For example, AMOC strength can be derived from characteristic patterns in North Atlantic sea surface temperature (SST). Due to the AMOC transporting large quantities of heat northwards, a weakening of the circulation leads to decreased heat transport to the subpolar North Atlantic, resulting in a distinctive pattern consisting of cooling south of Greenland and warming along the Gulf Stream region of the western Atlantic, as shown in figure 2. Caesar et al. demonstrated that this fingerprint is present in observational SST datasets and closely resembles that produced by climate simulations of the AMOC weakening. Working on this relationship, it is estimated that the AMOC has weakened by approximately 3 ± 1 Sverdrups, equivalent to roughly 15% since the mid-twentieth century. Their reconstructed index also indicates that recent decades contain the weakest AMOC state of the observational record [7].

Oceanographic and paleoclimate proxy records also offer independent evidence of a weakening AMOC. Analyses of marine sediments, coral records and isotopic tracers suggest that present AMOC strength is unusually weak when compared with conditions during the last millennium. Proxy reconstructions indicate that, despite the AMOC naturally experiencing variability on decadal to centennial timescales, recent changes appear consistent with external forcing associated with anthropogenic climate change rather than internal variability alone. In addition, geological evidence demonstrates the AMOC's ability to shift between strong and weak modes during previous glacial periods, resulting in abrupt climatic transitions over short timescales. This historical evidence confirms that large changes in AMOC strength are possible within the Earth system [1, 5].

Climate model simulations provide further evidence of AMOC weakening as a response to anthropogenic climate change. While the magnitude of projected weakening varies considerably between models, almost

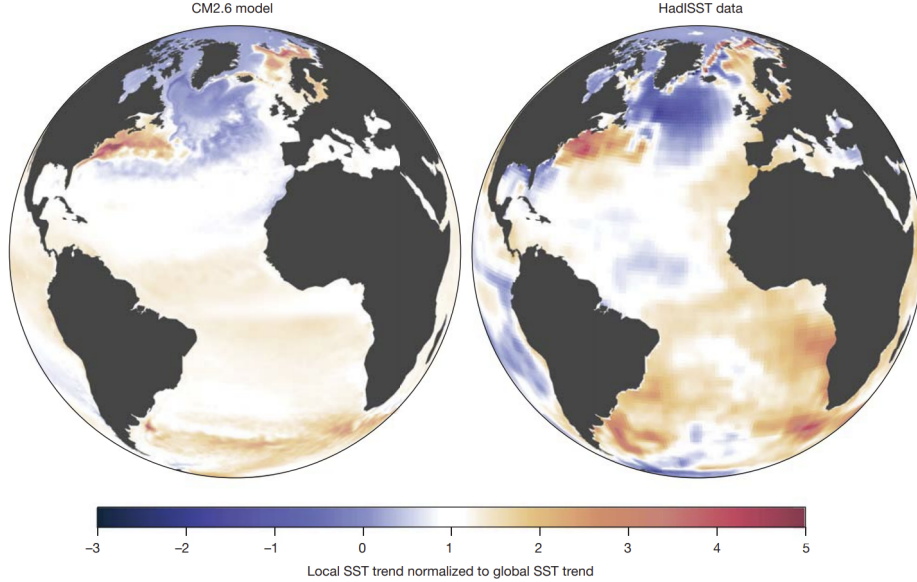


Figure 2: Modelled (left) and observed (right) North Atlantic sea surface temperature fingerprint associated with AMOC weakening. Cooling south of Greenland and warming along the Gulf Stream arise from reduced northward heat transport. Adapted from [7].

all CMIP5 and CMIP6 simulations predict a reduction in AMOC strength under increasing greenhouse gas concentrations during the twenty-first century. The Intergovernmental Panel on Climate Change (IPCC) Fifth Assessment Report concluded that the AMOC is *very likely* to weaken over the coming century under all major emissions scenarios, although a complete collapse before 2100 was assessed as *very unlikely* [3]. Despite this broad agreement, AMOC stability remains a contentious topic. More recent studies have suggested that many climate models underestimate the sensitivity of the AMOC as a result of biases in the representation of freshwater transport, ocean mixing, and deepwater formation and correcting these biases has been shown to fundamentally alter the simulated response to increasing CO_2 concentrations. Liu et al. demonstrated that, rather than exhibiting a gradual weakening, the corrected model eventually experienced a complete AMOC collapse, highlighting that model biases may underestimate true AMOC sensitivity [4, 8].

Similarly, North Atlantic convection studies have implied that many climate models underestimate the likelihood of abrupt regional cooling due to reduced deep water formation. Climate models capable of more realistic North Atlantic stratification reproduction were significantly more likely to experience abrupt cooling associated with weak convection with results implying that current models, less suited to ocean stratification and deep convection simulation, may underestimate the probability of rapid changes in the North Atlantic [8].

Owing to the AMOC's far reaching affects, the implications of continued AMOC weakening extend far beyond the North Atlantic. Simulations indicate that, beyond a cooling across the North Atlantic, a substantial reduction in AMOC strength would lead to changes in atmospheric circulation, southward displacement of tropical rainfall belts, altered storm tracks and significant impacts on European temperature and precipitation. Further consequences include regional sea level rise along the eastern coast of North America, reduced marine life productivity and potential interactions with other tipping elements such as the Greenland Ice Sheet and Amazon rainforest [2, 9].

Although the exact future evolution of the AMOC remains uncertain, the collective evidence from observations, reanalysis, proxy records and climate models strongly suggest a weakening circulation, with recent studies indicating that the present day AMOC may be exhibiting characteristics associated with declining stability and increasing susceptibility to abrupt transition. Detecting reliable early warning signals capable of identifying these changes before a critical threshold is reached therefore represents a vital challenge in climate science and one that forms the central motivation for this study.

1.3 Tipping Points

While many components of the climate system respond in an almost linear fashion to gradual external forcing, some systems can exhibit strong non-linear behaviour, wherein small changes in forcing can cause large changes in the system's state once a critical threshold is reached. These thresholds are known as *climate tipping points*, with components of the climate system that are capable of displaying this behaviour being known as *climate tipping elements* [10, 17].

Tipping points can be understood as critical thresholds, beyond which small perturbations are capable of altering the state of a system [17]. Before these thresholds are reached, negative feedback mechanisms typically act to return the system to its previous equilibrium after experiencing a perturbation. However, as the system loses stability, these mechanisms weaken. Once a threshold is crossed, the transition may become self-sustaining, owing to the strengthening of internal positive feedbacks, allowing the system to rapidly change state even if external forces are gradual.

One way to conceptualise this behaviour is by considering a bistable system, that is one with two stable equilibrium states separated by one unstable state. Under stable conditions, the system recovers quickly from perturbations away from the current equilibrium. As the stability of the system is affected by external forcing the equilibrium gradually becomes less resistant to disturbance until a critical point is reached, beyond which the original stable state ceases to exist and the system is forced to transition to the other equilibrium. This behaviour is associated with hysteresis, meaning that stopping or even reversing the forcing may not be sufficient to restore the system to its previous state [11].

A number of components of Earth's climate system have been identified as potential tipping elements, such as the Greenland and West Antarctic Ice Sheets, the Amazon rainforest, permafrost regions and the AMOC [10, 17]. The principal climate tipping elements identified by Lenton et al. are shown in Figure 3. More recent assessments have created a distinction between global 'core' tipping elements, which have substantial effects on the functioning of the global climate system, and regional 'impact' tipping elements, whose transitions would have significant regional environmental or societal consequences [17]. While the physical processes that control each element differ, they share the possibility of gradual anthropogenic forcing causing abrupt and potentially irreversible responses.

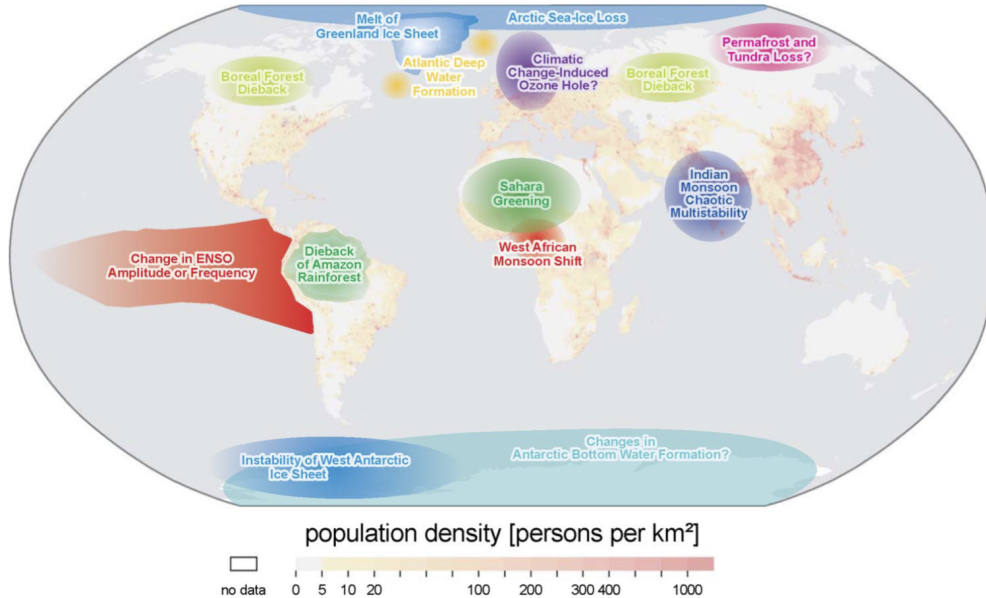


Figure 3: Major climate tipping elements identified within the Earth system. Continued anthropogenic warming has the potential to destabilise several interacting components, including the Atlantic Meridional Overturning Circulation (AMOC). Adapted from [10].

Future climate projections are further complicated by complex interactions between different tipping elements. The transition of one element may affect the forcing of another, potentially increasing the proba-

bility of additional transitions and, in extreme cases, producing a cascade of interacting tipping events [17]. Uncertainties regarding the magnitude of these interactions mean that individual tipping elements cannot necessarily be considered entirely independently of one another.

The existence of these tipping points presents a crucial challenge for traditional climate prediction reliant on conventional trend analysis, which is primarily concerned with changes in the mean state of a system. However, a system approaching a tipping point may not exhibit significant change in its mean state until shortly before experiencing an abrupt transition. Therefore, the detection of gradual trends alone may not provide sufficient information regarding changes in the underlying stability of a system. Instead, attention should be given to *early warning signals*, which aim to identify the approach towards a critical transition before the transition itself occurs.

1.4 Early Warning Signals

Early warning signals (EWS) are measurable changes in the behaviour of a system indicative of declining stability and, therefore, increasing susceptibility to abrupt transition. These indicators attempt to quantify the gradual reduction of resilience that may occur as a system approaches a critical threshold, as opposed to identifying a tipping point directly. Figure 4 illustrates the behaviour of commonly used statistical early warning indicators for systems with and without a critical transition [5].

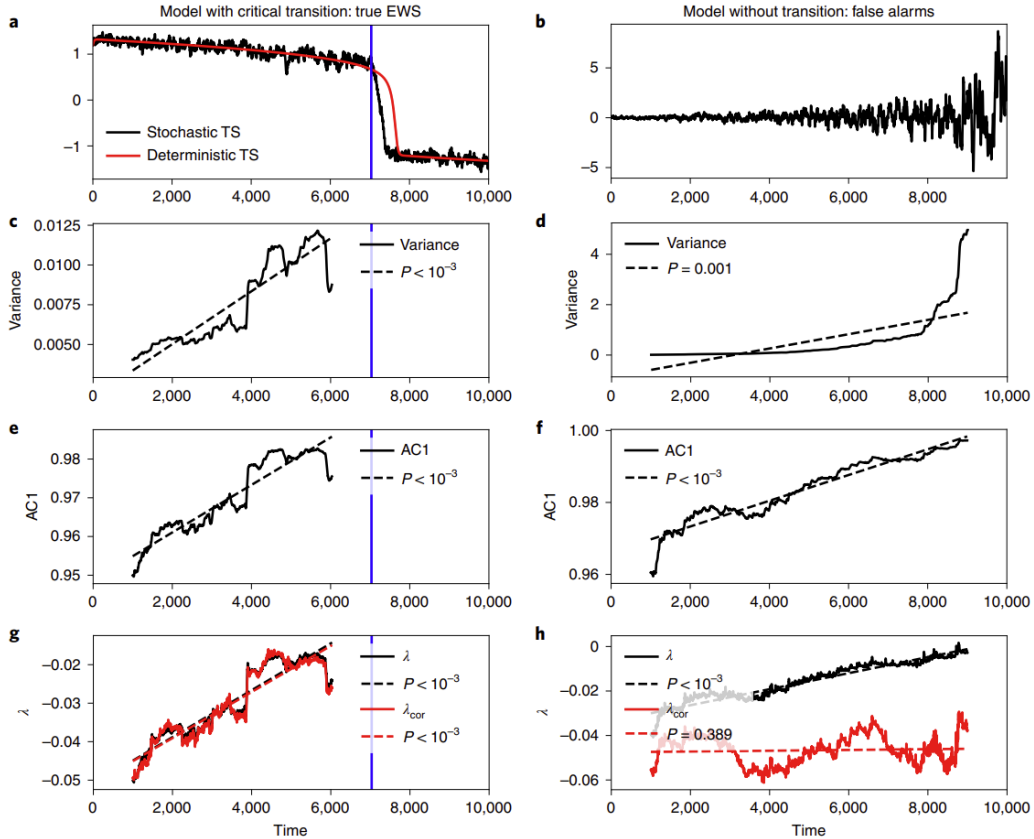


Figure 4: Comparison of several commonly used early warning signal indicators. The left column shows the behaviour of each indicator for a system approaching a genuine critical transition while the right column illustrates a system without a transition in which increasing variance and autocorrelation produce false alarms. Adapted from [5].

Systems such as the AMOC, with limited observational data and with little known regarding a critical threshold, are particularly suited to the use of EWS. If changes in climate variables contain information

regarding AMOC stability, these variables could provide evidence of destabilisation before large scale reorganisation occurs. Therefore, both statistical and physics-based early warning signals have been investigated using both observational data and climate models [5, 6].

1.4.1 Critical Slowing Down and Statistical Indicators

One of the most studied precursors of a critical transition is known as *critical slowing down*. As mentioned prior, a stable system responds to perturbation by returning to its equilibrium state and as the system approaches a tipping point its restoring forces gradually weaken, increasing the time required to recover from a disturbance. The system therefore retains a memory of previous perturbations, providing a measurable indication of its weakening [5, 11].

Characteristic changes in the statistical properties of a time series can arise from a critical slowing down, with one commonly used indicator being increasing lag-1 autocorrelation. Autocorrelation is a measure of the similarity between neighbouring observations in time. If a system recovers rapidly from perturbations, the relations between consecutive deviations from equilibrium are weak. However, as recovery slows the effect of each perturbation lasts longer and neighbouring observations become increasingly correlated. An increase in lag-1 autocorrelation may be an indicator for a reduction in the restoring ability of the system [5, 11].

Increasing variance is another commonly used indicator. As system stability weakens, perturbations are damped less effectively and larger deviations from the equilibrium state can occur. The result is an increase in distribution width of system states and, therefore, an increase in the variance of the corresponding time series. Consequently, increasing variance and autocorrelation have seen extensive application as statistical early warning signals of critical transitions [5, 11].

These indicators, however, are not without limitations. Changes in variance or autocorrelation may arise due to changes in the statistical properties of the external forcing as opposed to the underlying system itself becoming less stable, creating the possibility of false early warning signals. To counter this, Boers introduced an additional measure based on the restoring rate, λ , obtained via the linearisation of the dynamics of the system around its stable equilibrium. For a stable state, λ is negative, while a value approaching zero indicates progressively slower recovery from perturbations. Boers suggests that the approach of λ towards zero can provide a more reliable estimate of changing stability than variance alone [5].

Through the application of these methods to eight independent AMOC indices constructed from sea surface temperature and salinity observations in the Atlantic, Boers identified significant and spatially consistent early warning signals across the Atlantic basin. The results suggest that the AMOC underwent a substantial loss of stability during the twentieth century and may have, therefore, moved closer to a critical transition [5]. While these results do not determine when an AMOC transition will occur, they do demonstrate that information regarding circulation resilience can be derived from observable climate variables.

1.4.2 Physics-Based Early Warning Signals

Statistical early warning signals infer changes in stability from the behaviour of climate time series without necessarily requiring direct knowledge of the physical mechanism driving the transition. An alternative approach is to construct early warning indicators from the underlying physical mechanisms controlling AMOC stability.

One such physical mechanism governing AMOC stability is the transport of freshwater through the Atlantic Ocean. The transport of salt and freshwater between the Atlantic and surrounding ocean basins is heavily affected by changes in overturning strength. This relationship can cause a feedback in which an initially weaker AMOC causes additional freshening of the North Atlantic surface waters, further reducing their density and resulting in additional weakening of the AMOC [4, 6]. The freshwater transport associated with the AMOC can therefore provide information regarding its stability.

This relationship was investigated by Van Westen et al. using long duration simulation of the community Earth System Model, in which freshwater forcing over the North Atlantic was subject to a gradual increase until the AMOC exhibited an abrupt transition. From this experiment, a physics-based early warning signal was developed based on the minimum in AMOC induced freshwater transport across the southern boundary of the Atlantic. This indicator changed systematically before AMOC collapse and was less dependent on the choice of statistical analysis window than traditional variance and autocorrelation indicators [6].

When applied to ocean reanalysis products, the same indicator’s observed trend was interpreted as evidence that the present day AMOC may be approaching a tipping point [6]. While, as with statistical indicators this does not provide a precise prediction for when a future transition may occur, it demonstrates that changes in physically meaningful ocean variables may provide additional information regarding AMOC stability.

The statistical and physics-based approaches are complimentary: statistical indicators can reveal changes in the behaviour of a system without requiring a complete physical model, while observed changes can be connected directly to processes known to influence AMOC stability by physics-based indicators. A combination of multiple climate variables and forms of early warning signals may therefore provide a more robust method of identifying changes in the AMOC than any single indicator can. This motivates the use of machine learning methods which are capable of identifying complex non linear and multivariate relationships within large spatiotemporal climate datasets

1.5 Machine Learning for Climate Prediction

Climate datasets possess an inherent multidimensionality, non-linearity and strong spatial and temporal correlations. While traditional statistical methods remain essential for the identification of trends, they may struggle with the representation of complex relationships between many interacting variables. Machine learning provides a complementary approach in which relationships between variables can be inferred from the data rather than being specified in advance.

Machine learning refers to a broad range of computational methods that learn from data in order to complete tasks such as classification and regression. These methods can be applied to large observational, reanalysis and model datasets containing multiple variables with both spatial and temporal dimensions. This makes machine learning particularly suited to problems without a strong trend in a single variable but rather a signal arising from non-linear interactions between multiple variables.

There exists an important distinction between spatial and temporal information in climate science; climate variables are spatially structured due to neighbouring grid cells being physically related, while successive measurements are temporally correlated owing to persistence, seasonal cycles and long term climate variability. An ideal forecasting model should therefore be able to learn both types of dependence. Convolutional Neural Networks (CNNs) are useful for the recognition of spatial structure, while methods like Long Short-Term Memory (LSTM) networks are used for the representation of temporal dependence. These two approaches can be combined to form Convolutional LSTM networks which are well suited to spatiotemporal climate forecasting [13].

1.5.1 Classification

Classification is a supervised machine learning method in which a model learns to assign input data to one of a set of predefined categories. The model is trained on example data for which the correct category is known and learns to map between the input features and the associated class label. Once trained, the model can be applied to unseen data, for which the class labels are unknown.

In the context of climate science, classification serves as one possible method of identifying differences between climate regimes. For example, with climate data divided into several periods representing different background states, a model can be trained to determine whether these periods can be distinguished from their associated physical variables. If the statistical properties of each period are similar, classification becomes difficult. On the other hand, high classification accuracy may indicate a systematic change in the underlying data between periods.

A subtle change in climate distributed across temperature, salinity, ocean currents and atmospheric quantities, which may be difficult to identify using independent univariate trends, can be detected by classifiers being able to combine information from multiple variables simultaneously means that

Subtle changes in temperature, salinity, ocean currents and atmospheric variables may be difficult to identify using independent univariate trends, but together may produce a distinguishable change in the state of the system as a whole. Where traditional statistical methods fall short, machine learning classifiers may detect this change in the state of the system; the potential of this approach has been demonstrated via the application of classifiers to distinguish meteorological data from different historical climate periods [14].

However, classification does not provide a direct prediction of future climate evolution. Regression and forecasting models are required for this purpose, with deep neural network architectures being particularly useful when the data has both spatial and temporal elements, motivating the use of CNN and ConvLSTM models in climate forecasting.

1.5.2 Convolutional Neural Networks

Convolutional Neural Networks (CNNs), originally developed for the analysis of spatially structured data, apply small filters across local regions of an input array instead of connecting every input to every neuron directly. By applying the same filter across the full spatial domain, the network can identify patterns irrespective of location and significantly reduce the number of trainable parameters when compared to a fully connected network [12,14]. CNNs are therefore well suited to climate data, where neighbouring spatial locations are physically related and patterns often extend across multiple grid cells [14].

For two-dimensional climate variables, filters move across the latitudinal and longitudinal dimensions and a weighted combination of neighbouring grid cells is calculated. The weights of these filters are adjusted such that the network learns spatial features relevant to the prediction task, with the early convolutional layers identifying relatively simple local gradients or boundaries and deeper layers combining these features into increasingly complex spatial patterns [12]. When applied to climate data, this allows CNNs to extract spatial structures like temperature gradients directly from the gridded data [14].

This makes CNNs particularly useful when applied to atmospheric and oceanographic datasets due to these systems being spatially continuous. Conditions at one grid cell are inherently related to those at nearby cells and structures such as fronts, gyres and temperature gradients extend across multiple grid cells. CNNs can make use of information from neighbouring location when making predictions as they preserve this spatial arrangement [13,14].

As powerful as CNNs are when applied to spatial data, they do not contain explicit mechanisms for the handling of temporal information. Climate forecasting is intrinsically spatiotemporal, as the future state of a system is reliant upon not only its spatial structure but also its previous evolution, and, although a CNN can handle previous time steps provided as input, recurrent neural network architectures are capable of representing temporal dependence in a more direct way. This limitation of CNNs drives the use of Long Short-Term memory networks and, ultimately, Convolutional LSTM architectures that combine both temporal and spatial handling abilities [13].

1.6 Long Short-Term Memory Networks

Recurrent Neural Networks (RNNs) are designed for sequential data but differ from conventional methods by maintaining information about previous inputs. This enables the network to use earlier states of a sequence while processing later states, making RNNs suitable for time-series forecasting [14,15].

Traditional recurrent networks may struggle when learning dependencies across long sequences as gradients can become increasingly small or large in training. To reduce this problem, Long Short-Term Memory (LSTM) networks were developed which introduce a persistent internal cell state along with a series of gates to control the flow of information through the network [15].

Typically, an LSTM cell contains three gates: an input gate, which controls what new information should be added, a forget gate, which determines which information from the previous cell should be retained, and an output gate, which determines which part of the updated cell state contributes to the hidden state passed to the next time step. This architecture allows for useful information to persist over longer sequences while irrelevant information is discarded [15].

This ability to retain temporal information is particularly useful for climate data as many physical processes have dependencies that span over multiple timescales. The state of the climate at a given moment in time cannot be considered independently of preceding conditions owing to seasonal cycles, ocean heat storage, and persistent circulation anomalies, this makes the ability of LSTMs to learn temporal relationships that purely convolutional networks cannot a powerful tool for climate forecasting [13,14].

However, standard LSTM networks operate on one-dimensional feature vectors, meaning that latitude and longitude fields must generally be flattened before processing. This removes the spatial relationships between neighbouring grid cells and limits the application of conventional LSTMs to spatially gridded data [13].

1.6.1 Convolutional LSTM Networks

Convolutional Long Short-Term Memory networks (ConvLSTMs) were developed as an extension of standard LSTMs to include spatial sequence prediction [13]. Fundamentally, ConvLSTMs differ from standard LSTMs by replacing the matrix multiplications with convolution operations, maintaining the spatial grid form of hidden and cell states as opposed to converting them to one-dimensional vectors. This allows the algorithm to preserve spatial relationships between neighbouring grid cells whilst simultaneously learning temporal dependencies [13]. The internal structure of a ConvLSTM cell is illustrated in Figure 5.

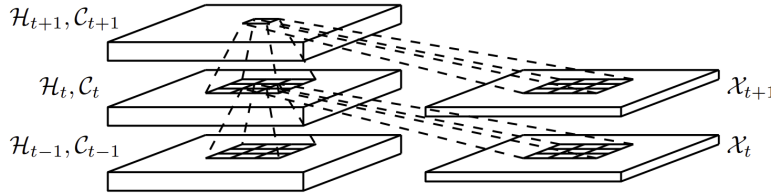


Figure 5: Internal structure of a Convolutional Long Short-Term Memory (ConvLSTM) cell. The input tensor, previous hidden state and previous cell state are combined through convolutional operations to produce updated hidden and cell states while preserving the spatial structure of the data. Adapted from [13].

Therefore, a ConvLSTM combines the temporal memory of an LSTM with the spatial feature handling of a CNN. The model receives a spatial field for each time step and combines it with the information retained from previous time steps. The algorithm learns spatial relationships between neighbouring grid cells due to these convolutional operations while relevant information is carried forwards through the sequence by the recurrent cell state, allowing the network to model both spatial and temporal dependencies simultaneously [13].

These spatiotemporal capabilities make ConvLSTM networks particularly well suited to climate forecasting where, generally, input data consists of sequences of two-dimensional spatial fields. For example, variables such as sea surface temperature, salinity and ocean currents exhibit both spatial coherence and temporal persistence. ConvLSTMs are ideal for this kind of data as they are capable of learning spatiotemporal relationships directly without requiring the flattening of the spatial dimensions into one-dimensional vectors [13, 14].

Previous work at Royal Holloway applied CNN, LSTM and ConvLSTM algorithms for monthly gridded meteorological forecasting, finding that the standalone LSTM performed poorly when the spatial grid was flattened while the CNN performed substantially better for spatial forecasting in a single step. By combining the strengths of both approaches, the ConvLSTM achieved significantly stronger forecasting performance over longer periods, highlighting the importance of taking both spatial and temporal dependencies into account for climate modelling [14].

The objective of the present study, however, is not to reproduce future climate variables as accurately as possible. A model trained on 'historical' climate data (pre modern anthropogenic forcing) learns the spatiotemporal relationships of that climate state and uses this information to predict future variables. Disagreement between observed measurements and the model predictions would suggest an evolution in those relationships, providing information regarding changes in the underlying climate system. This motivates the investigation of whether ConvLSTM models can be used not only for forecasting but for the detection of early warning signals of a weakening AMOC.

1.7 Research Objectives

The primary objective of this study is to investigate the use of machine learning as an anomaly detection tool for the identification of early warning signals of AMOC weakening. The study examines whether changes in the effectiveness of climate prediction methods applied to ocean and atmospheric variables can reveal changes in AMOC behaviour, as opposed to attempting to estimate AMOC strength directly.

To achieve this, the study is divided into three stages. Firstly, an inflection point within historical climate data is identified, corresponding to the introduction of anthropogenic forcing. This inflection point is found using statistical methods, including piecewise linear regression and supervised classification, and separates 'historical' data (used later for model training) from the modern climate period.

Secondly, this 'historical', pre-inflection-year data is then used to train a ConvLSTM network which is then used to forecast the subsequent evolution of multiple ocean and atmospheric variables over the North Atlantic area.

Lastly, these predicted climate variables are compared with observational measurements from ocean and atmospheric reanalysis datasets. The underlying hypothesis of this study is that a ConvLSTM trained on historical climate data should make accurate predictions of conditions that continue to evolve according to the physical relationships of a historical climate state. If, however, these relationships have changed due to anthropogenic forcing, the divergence of the predicted and observed data should increase systematically through time. This prediction error can therefore be treated as an anomaly detection metric that may provide evidence for gradual changes in climate system behaviour.

This analysis is repeated for multiple climate variables and geographical regions selected from literature as being physically relevant to AMOC stability. The application of machine learning to the identification of early warning signals of AMOC weakening is investigated by the comparison of evolving prediction errors across these variables.

The study therefore aims to address the following questions:

1. Can an objective inflection point separating historical and modern climate states be identified using statistical and machine learning techniques?
2. Can a ConvLSTM model trained exclusively on the historical climate state accurately reproduce the subsequent evolution of North Atlantic climate variables?
3. Does prediction error increase systematically following the identified inflection point, consistent with a changing climate?
4. Which oceanic and atmospheric variables exhibit the strongest anomaly signals and do these correspond to regions identified as being indicative of AMOC stability?
5. Can ConvLSTM anomaly detection provide a complementary method for the identification of potential early warning signals of a weakening AMOC?

2 Datasets and Methodology

2.1 Study Region

2.2 Datasets

2.2.1 ORAS5 Ocean Reanalysis

2.2.2 ERA5 Atmospheric Reanalysis

2.2.3 Variables Used

2.3 Data Pre-processing

2.3.1 Spatial Subsetting

2.3.2 Re-gridding and Combination

2.3.3 Normalisation

2.3.4 Training, Validation and Test Sets

2.4 Inflection Point Study

2.5 ConvLSTM Architecture

2.6 Model Training

2.7 Performance Metrics

2.8 Early Warning Signal Analysis

3 Model Performance

4 Conclusion

A Additional Figures

References

- [1] Rahmstorf, S. (2002). Ocean circulation and climate during the past 120,000 years. *Nature*, 419(6903), 207–214.
- [2] Jackson, L. C., Kahana, R., Graham, T., Ringer, M. A., Woollings, T., Mecking, J. V., & Wood, R. A. (2015). Global and European climate impacts of a slowdown of the AMOC in a high resolution GCM. *Climate Dynamics*, 45, 3299–3316.
- [3] Collins, M., Knutti, R., Arblaster, J., Dufresne, J.-L., Fichet, T., Friedlingstein, P., et al. (2013). Long-term Climate Change: Projections, Commitments and Irreversibility. In: *Climate Change 2013: The Physical Science Basis*. Cambridge University Press.
- [4] Liu, W., Xie, S.-P., Liu, Z., & Zhu, J. (2017). Overlooked possibility of a collapsed Atlantic Meridional Overturning Circulation in warming climate. *Science Advances*, 3(1), e1601666.
- [5] Boers, N. (2021). Observation-based early-warning signals for a collapse of the Atlantic Meridional Overturning Circulation. *Nature Climate Change*, 11, 680–688.
- [6] van Westen, R. M., Kliphuis, M., & Dijkstra, H. A. (2024). Physics-based early warning signal shows that AMOC is on tipping course. *Science Advances*, 10(6), eadk1189.

- [7] Caesar, L., Rahmstorf, S., Robinson, A., Feulner, G., & Saba, V. (2018). Observed fingerprint of a weakening Atlantic Ocean overturning circulation. *Nature*, 556, 191–196.
- [8] Sgubin, G., Swingedouw, D., Drijfhout, S., Mary, Y., & Bennabi, A. (2017). Abrupt cooling over the North Atlantic in modern climate models. *Nature Communications*, 8, 14375.
- [9] Drijfhout, S. (2015). Competition between global warming and an abrupt collapse of the AMOC in Earth’s energy imbalance. *Scientific Reports*, 5, 14877.
- [10] Lenton, T. M., Held, H., Kriegler, E., Hall, J. W., Lucht, W., Rahmstorf, S., & Schellnhuber, H. J. (2008). Tipping elements in the Earth’s climate system. *Proceedings of the National Academy of Sciences*, 105(6), 1786–1793.
- [11] Scheffer, M., Bascompte, J., Brock, W. A., Brovkin, V., Carpenter, S. R., Dakos, V., Held, H., van Nes, E. H., Rietkerk, M., & Sugihara, G. (2009). Early-warning signals for critical transitions. *Nature*, 461, 53–59.
- [12] LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278–2324.
- [13] Shi, X., Chen, Z., Wang, H., Yeung, D.-Y., Wong, W.-K., & Woo, W.-C. (2015). Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting. In *Advances in Neural Information Processing Systems*, 28.
- [14] Boyce, I. (2024). *Applications of Machine Learning in Climate Science*. MSc by Research Thesis, Royal Holloway, University of London.
- [15] Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, 9(8), 1735–1780.
- [16] Keating, C. (2026). *AMOC: Is global warming tipping key Atlantic ocean currents towards “collapse”?* Carbon Brief. Available at: <https://interactive.carbonbrief.org/amoc-explainer/index.html>
- [17] Armstrong McKay, D. I., Staal, A., Abrams, J. F., Winkelmann, R., Sakschewski, B., Loriani, S., Fetzer, I., Cornell, S. E., Rockström, J., & Lenton, T. M. (2022). Exceeding 1.5°C global warming could trigger multiple climate tipping points. *Science*, 377(6611), eabn7950.